

Evaluation of a Deep Learning System For Identifying Glaucomatous Optic Neuropathy Based on Color Fundus Photographs

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Precis: Pegasus outperformed 5 of the 6 ophthalmologists in terms of diagnostic performance, and there was no statistically significant difference between the deep learning system and the “best case” consensus between the ophthalmologists. The agreement between Pegasus and gold standard was 0.715, whereas the highest ophthalmologist agreement with the gold standard was 0.613. Furthermore, the high sensitivity of Pegasus makes it a valuable tool for screening patients with glaucomatous optic neuropathy.

Purpose: The purpose of this study was to evaluate the performance of a deep learning system for the identification of glaucomatous optic neuropathy.

Materials and Methods: Six ophthalmologists and the deep learning system, Pegasus, graded 110 color fundus photographs in this retrospective single-center study. Patient images were randomly sampled from the Singapore Malay Eye Study. Ophthalmologists and Pegasus were compared with each other and to the original clinical diagnosis given by the Singapore Malay Eye Study, which was defined as the gold standard. Pegasus’ performance was compared with the “best case” consensus scenario, which was the combination of ophthalmologists whose consensus opinion most closely matched the gold standard. The performance of the ophthalmologists and Pegasus, at the binary classification of nonglaucoma versus glaucoma from fundus photographs, was assessed in terms of sensitivity, specificity and the area under the receiver operating characteristic curve (AUROC), and the intraobserver and interobserver agreements were determined.

Results: Pegasus achieved an AUROC of 92.6% compared with ophthalmologist AUROCs that ranged from 69.6% to 84.9% and the “best case” consensus scenario AUROC of 89.1%. Pegasus had

a sensitivity of 83.7% and a specificity of 88.2%, whereas the ophthalmologists’ sensitivity ranged from 61.3% to 81.6% and specificity ranged from 80.0% to 94.1%. The agreement between Pegasus and gold standard was 0.715, whereas the highest ophthalmologist agreement with the gold standard was 0.613. Intraobserver agreement ranged from 0.62 to 0.97 for ophthalmologists and was perfect (1.00) for Pegasus. The deep learning system took ~10% of the time of the ophthalmologists in determining classification.

Conclusions: Pegasus outperformed 5 of the 6 ophthalmologists in terms of diagnostic performance, and there was no statistically significant difference between the deep learning system and the “best case” consensus between the ophthalmologists. The high sensitivity of Pegasus makes it a valuable tool for screening patients with glaucomatous optic neuropathy. Future work will extend this study to a larger sample of patients.

Key Words: Artificial Intelligence (AI), deep learning, optic neuropathy, online retinal fundus image database for glaucoma analysis and research (ORIGA), Singapore Malay Eye Study (SiMES)

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Glaucoma is the leading cause of irreversible blindness in the world.¹ Its prevalence was ~64 million in 2013, with projections of an increase to 76 million by 2020, and 111.8 million by 2040.¹ Although the prevalence of glaucoma is predicted to increase, the number of ophthalmologists has been stable over the years.¹ Additional tools are required to improve the efficiency of glaucoma screening. Recently, efforts have been made to develop Artificial Intelligence (AI) tools for this purpose.¹

Deep learning is a subfield of AI. Deep learning algorithms used in computer vision are based on a type of artificial neural network known as Convolutional Neural Networks (CNNs).^{2–4} CNNs are loosely inspired by biology; they consist of a number of units called neurons that are connected together. Each neuron performs simple operations on its inputs and then feeds the result into connected neurons. The neurons are often connected in a layered hierarchy, and each neuron has a small number of parameters that can be tuned (learned) so that the aggregate system of neurons is able to perform simple tasks. Deep learning refers to having many layers of neurons, allowing the algorithm to learn a hierarchy of features in the data that are useful for classification tasks such as whether a patient suffers from glaucoma based on a photograph of their optic disc. Deep learning algorithms have achieved state-of-the-art performance for a number of visual tasks, and thus there has been a recent effort to apply deep

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learning to health care. Recently, several deep learning algorithms have been developed to detect glaucomatous optic neuropathy based on fundus photography.²⁻⁴

Pegasus (Visulytix Ltd, London, UK) is an AI system that utilizes deep learning to autonomously screen for glaucomatous optic neuropathy by recognizing and interpreting features within fundus photography. The purpose of this study is to evaluate the performance of Pegasus and to determine its efficacy as an additional tool to be used in glaucoma screening.

MATERIALS AND METHODS

This study adhered to the tenets of the Declaration of Helsinki and was approved by the Columbia University Institutional Review Board, which determined that the study qualified for a waiver of written informed consent.

Six ophthalmologists, with varying levels of experience ranging from residents and glaucoma fellows in training to specialists in glaucoma and neuro-ophthalmology, evaluated 110 macula-centered fundus photographs that were randomly selected from the publicly available Online Retinal fundus Image database for Glaucoma Analysis and research (ORIGA).^{5,6} The ORIGA dataset was originally collected as part of the Singapore Malay Eye Study (SiMES).⁷ SiMES was conducted from 2004 to 2007 by the Singapore Eye Research Institute and examined 3280 Malay adults aged 40 to 79 years, from which, 150 patients were diagnosed as having glaucoma. The ORIGA dataset was constructed by taking 168 glaucomatous images and randomly sampling 482 nonglaucoma images from SiMES. Of the glaucoma images, it is expected that 69% have primary open angle glaucoma, 5% have primary angle-closure glaucoma, and 18% have secondary glaucoma.⁷ Of the nonglaucoma images, patients could suffer from other diseases such as myopia (defined as spherical equivalent refraction of < -0.5 D), age-related macular degeneration, diabetic retinopathy, ocular trauma, or retinal vein occlusion.⁷ The prevalence of these diseases within the dataset range from 0.7% in the case of retinal vein occlusion, to 5.6% in the case of age-related macular degeneration.⁷ The prevalence of myopia within the study population was 24.6%. All patients were of Malay ethnicity with a mean age of 58 ± 11 years. The characteristics of the subsample of images used in this evaluation are summarized in Table S1 (Supplemental Digital Content 1, <http://links.lww.com/IJG/A292>).

In this study, 110 images were sampled from the ORIGA dataset, of which 10 were duplicates (5 glaucoma, 5 nonglaucoma) for testing intraobserver reliability. The remaining 100 images were randomly sampled such that 50 had a glaucoma diagnosis and 50 were nonglaucomatous controls, based on the original diagnosis from SiMES. All images involved in this study were considered clinically gradable with minor artifacts. The 4 ophthalmologists (2 glaucoma fellows, glaucoma specialist, and neuro-ophthalmologist) and 2 members in training (second-year and third-year resident) graded each image without knowledge of the clinical diagnosis, or the presence of the duplicate images. The ophthalmologists were asked to assign confidence (ranging from 1 to 7) that glaucoma was present. This decision was based upon several factors of the fundus image including vertical cup to disc ratio (VCDR), disc diameter, disc hemorrhage, and retinal nerve fiber layer defect. Using a confidence-based grading system allowed for the correction of bias when evaluating and comparing performance and for a range of sensitivity and specificity values

to be determined for each grader depending on a confidence threshold. Glaucoma was considered to be a VCDR of ≥ 0.7 with an associated functional visual loss on perimetry, or VCDR of ≥ 0.6 with a disc notch.

The raw 3072-by-2048 pixel lossy JPEG images without preprocessing were manually uploaded to a cloud-based instance of the Pegasus deep learning system provided by Visulytix Ltd.

Pegasus is a clinical decision support software for the screening of retinal imagery, including fundus photographs and optical coherence tomography volumes. For fundus photography, the software utilizes a collection of CNNs, each specializing in a different task associated with image evaluation, including region detection (eg, optic disc, macula and fovea); clinical feature detection (eg, exudates and hemorrhages); classification of abnormalities (eg, of the optic disc or macula); and pathology classification and grading (eg, diabetic retinopathy).

In this study, the performance of the CNNs responsible for optic disc abnormality detection was evaluated. This component is designed to generalize to any colored fundus photography that contains an optic disc and relies on a CNN to first extract and standardize the optic disc region before it is fed into a CNN which can classify the optic disc as being either clinically “normal” or “abnormal.” The output of this second algorithm is evaluated in this study, with optic discs flagged as “abnormal” by the software taken as potentially glaucomatous, and those identified as “normal” taken as nonglaucomatous. This CNN is based on the ResNet-50 architecture of He et al.⁸ The input size to the classification network is 224-by-224 pixels. During the development of the classification CNN, the network was trained from scratch with all parameters randomly initialized at the start of training. The optic disc extraction algorithm was developed using TensorFlow, v1.3.0, and the optic disc abnormality classification algorithm was developed using Keras, v2.0.8 with a TensorFlow, v1.3.0 backend. The training of the classification algorithm was performed with a batch size of 32 and using the RMSProp optimizer with a learning rate of 0.0001.

In total, the development dataset consisted of 136,146 anonymized fundus images collected from multiple private sources that suitably reflects variation in observed real-world use cases: patient demographics, types of fundus camera (eg, manufacturer, fields of view ranging from 30 to 50 degrees, resolution, mydriatic and nonmydriatic), and acquisition artefacts (eg, dust and scratches). Images used in algorithm development were labeled by between 1 and 5 board-certified specialist ophthalmologists from the United Kingdom. No patients or images from ORIGA or the SiMES were used in the development of the evaluated algorithms, this includes as training or validation data during algorithm training, or as data for internal testing performed by the manufacturer.

The system performed the analysis in real-time and the disc anomaly output was recorded, which is a value ranging from 0 to 1 and can be interpreted as the confidence that the system has that an abnormality is present in the optic disc of the uploaded image. These abnormalities are not necessarily indicative of glaucoma and can be due to the presence of other pathologies of the optic nerve head. The deep learning system was able to provide analyses for all uploaded images; none failed due to image quality, system failures, or other issues. All human graders were masked to the output of the deep learning system.

Both the human graders and the deep learning system were compared with the original clinical diagnosis from SiMES,⁷ which was taken as the gold standard. The motivation for this, over using the consensus of the 6 ophthalmologists, was 2-fold. First, as the gradings of each ophthalmologist would be involved in forming the consensus reference standard, measuring the performance of each ophthalmologist could be biased as the reference standard could be correlated with the gradings by individual ophthalmologists. In addition, it would not be possible to compare the deep learning system to the consensus opinion of ophthalmologists. Second, the SiMES clinical diagnosis used additional testing to determine the diagnosis and not just the appearance of the fundus photograph. These tests included Goldmann applanation tonometry, visual field tests, slit lamp biomicroscopy, optic disc imaging, and digital lens photography. We reason that these additional tests would allow for better identification of glaucoma, than using only the retinal photograph.⁹

The performance of the ophthalmologists and the deep learning system were assessed in terms of the sensitivity, specificity, and area under the receiver operating characteristic curve (AUROC). To remove bias from threshold selection, for all reported values of sensitivity and specificity, the confidence threshold was selected such that the binary outcome gave optimal agreement (Cohen κ coefficient) with the gold standard. In addition, the deep learning system performance was compared with the “best case” consensus scenario, which was determined by taking the mean average of the glaucoma gradings across the best 5 ophthalmologists. The confidence intervals (CIs) at the 95% significance level were determined using a bootstrap technique with 1000 replications, and the DeLong statistical test was used to test for significant difference between AUROCs.¹⁰ Statistical tests were not performed for comparing sensitivity and specificity as these values are dependent on the choice of threshold and are difficult to compare unless the cost of false positives and false negatives are considered.

For the subset of 10 photographs reviewed twice by each grader and the deep learning system, the intraobserver repeatability was measured using intraclass correlation coefficients calculated with 2-way mixed effects models to investigate the repeatability of the probability scores and VCDR estimates. intraclass correlation coefficients of <0.5, between 0.5 and 0.75, between 0.75 and 0.9, and >0.90 are indicative of poor, moderate, good, and excellent repeatability, respectively.¹¹

Statistical analyses were performed using commercially available software (Stata, version 14; StataCorp LP). Statistical significance was defined at P -value <0.05.

RESULTS

The AUROC, sensitivity, and specificity of individual graders, grader consensus, and the deep learning system are summarized in Table S2 (Supplemental Digital Content 1, <http://links.lww.com/IJG/A292>). Pegasus achieved an AUROC of 92.6% (95% CI: 88.2-96.2) compared with ophthalmologist AUROCs that ranged from 69.6% (95% CI: 60.0-78.8) to 84.9% (95% CI: 77.9-91.2) and the “best case” consensus scenario (excluding the neuro-ophthalmology specialist) AUROC of 89.1% (95% CI: 83.1-94.5). It was determined that Pegasus outperformed 5 of the 6 ophthalmologists with P -values ranging from 0.0001 to 0.019 using DeLong tests performed to compare The Pegasus AUROC to that of each grader and measured

against the gold standard. Pegasus performed better than the remaining ophthalmologist, 92.6% (95% CI: 88.2-96.2) versus 84.9% (95% CI: 77.9-91.2), but this result was not statistically significant ($P=0.078$).

The deep learning system obtained a sensitivity of 83.7% (95% CI: 74.5-92.0) and a specificity of 88.2% (95% CI: 80.4-95.6) at a confidence cutoff of 0.01, whereas the ophthalmologist sensitivity ranged from 61.3% (95% CI: 50.0-72.5) to 81.6% (95% CI: 72.1-90.4). Moreover, the specificity of Pegasus was 88.2 (95% CI: 80.4-95.6) compared with ophthalmologist specificity which ranged from 80.0 (95% CI: 70.8-89.1) to 94.2 (95% CI: 88.2-98.3).

When considering the “best case” consensus scenario, it was determined that the optimal scenario was 1 in which the worst performing grader (grader 6) was ignored, and the consensus was taken among the remaining 5 ophthalmologists. Figure 1 shows receiver operating characteristic curves comparing the sensitivity and specificity of the deep learning system to those yielded by this “best case” consensus scenario at a range of confidence thresholds. The “best case” consensus scenario yielded an AUROC of 89.1% (95% CI: 83.1-94.5), sensitivity of 89.8 (95% CI: 82.2-96.2), and specificity of 76.5 (95% CI: 66.7-86.0). In this case, the deep learning system performed better in terms of AUROC and specificity, but worse in terms of sensitivity. However, the difference between the deep learning system AUROC and “best case” consensus AUROC was not statistically significant ($P=0.346$).

Figure 2 shows the interobserver agreement matrix. The agreement between ophthalmologists was mostly moderate, with Cohen κ coefficients ranging from 0.45 to 0.61. The agreement between the deep learning system and ophthalmologists was also mostly moderate, with values ranging between 0.49 and 0.61. The agreement between the deep learning system and the gold standard was 0.715 (95% CI: 0.596-0.832), whereas the highest ophthalmologist agreement with the gold standard was 0.613 (95% CI: 0.479-0.737).

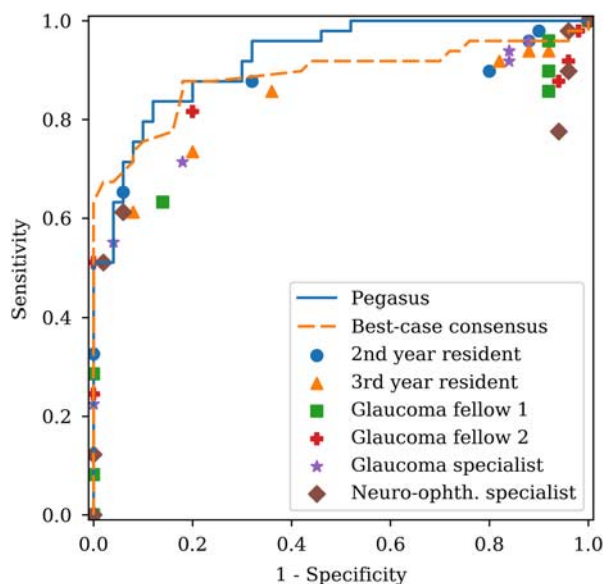


FIGURE 1. Receiver operating characteristic curves for the deep learning system (blue) and the “best case” consensus scenario between graders (dashed orange). The sensitivity and specificity for individual graders are plotted as points at a range of confidence thresholds.

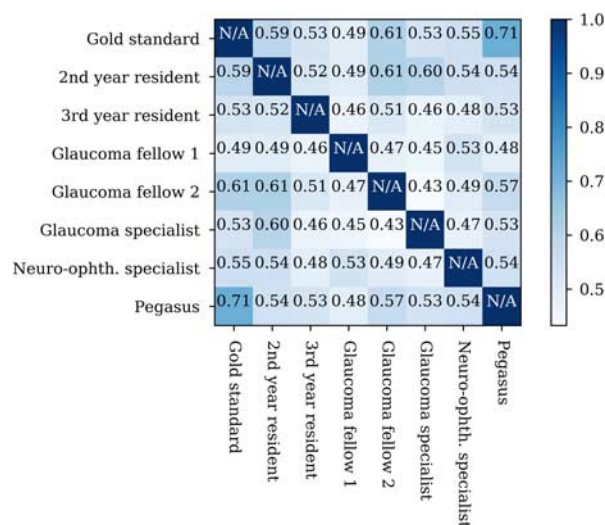


FIGURE 2. Interobserver agreement matrix. Agreement measured using Cohen κ coefficients. NA indicates not available.

Figures 3A–H and 4A–H show randomly selected examples of fundus photographs that were incorrectly and correctly classified by the deep learning system, respectively. Table S3 (Supplemental Digital Content 1, <http://links.lww.com/IJG/A292>) shows the corresponding grades for the 6 ophthalmologists. There are no obvious, common characteristics (biases) for the false-positive or false-negative groups. Of the false negatives (Figs. 3A–D), the consensus for 2 of the images (Fig. 3) were that they were borderline, whereas the consensus for the other 2 images were that they were glaucoma (in agreement with the gold standard and disagreement with the deep learning system), although the confidence was low. Of the false-positive examples (Figs. 3E–H), the consensus among ophthalmologists correctly identified that 1 image was nonglaucoma, whereas 2 were classified as borderline and the remaining image was incorrectly classified as glaucoma.

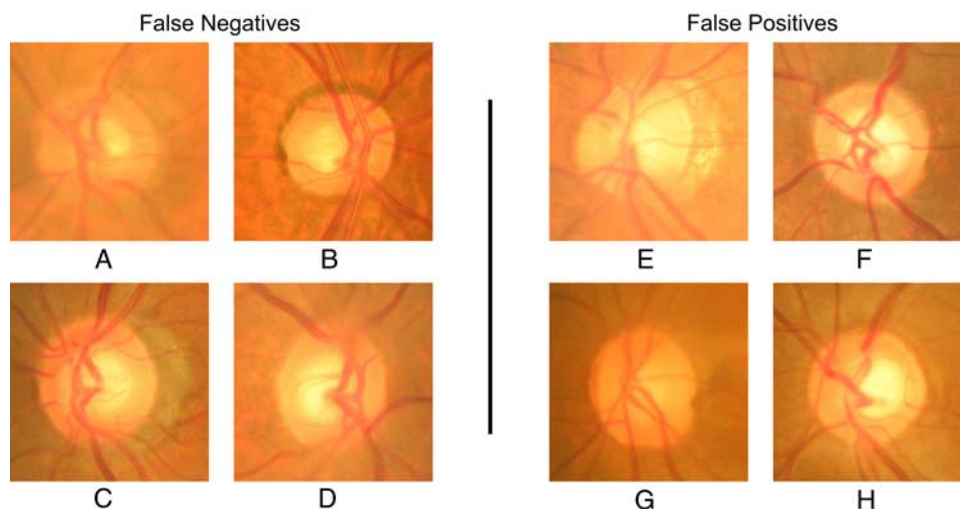


FIGURE 3. Randomly selected photographs that were classified incorrectly by the deep learning system. The false negatives (A–D) were identified as nonglaucoma by the deep learning system but were labeled as glaucoma in the gold standard, whereas the false positives (E–H) were identified as having glaucoma by the deep learning system but were labeled as nonglaucoma in the gold standard. The photographs have been manually cropped to just the optic disc region.

For the true-positive examples (Figs. 4A–D), the ophthalmologist consensus correctly identified 2 as glaucoma, whereas the other 2 images were classified as borderline. Of the true-negative examples, the consensus of ophthalmologists incorrectly identified 1 image as glaucoma, and the other 3 images were classified as borderline.

Table S4 (Supplemental Digital Content 1, <http://links.lww.com/IJG/A292>) summarizes the intraobserver agreement (repeatability) for both VCDR measurement and glaucoma diagnosis. For glaucoma, grader repeatability was good to excellent (range: 0.62 to 0.97). For VCDR, grader repeatability was excellent in all cases (range: 0.90 to 0.99). The Pegasus deep learning system had perfect repeatability for both the glaucoma diagnosis and VCDR.

Finally, the Pegasus deep learning system took on average 8 seconds to read and interpret an uploaded image and output glaucoma confidence and VCDR estimate, compared with human graders who took, on average, 90 seconds per image. Pegasus took a total of 14.7 minutes to grade the 110 images, compared with the average grader time of 2.25 hours. Therefore, Pegasus analyzed the images in ~10% of the time taken by a human grader.

DISCUSSION

In this study, Pegasus demonstrated a high sensitivity of 83.7% (95% CI: 74.5–92.0) and a high specificity of 88.2% (95% CI: 80.4–95.6) for the detection of glaucomatous optic discs from colored fundus photography. Moreover, Pegasus achieved a higher AUROC of 92.6% than the ophthalmologist graders when comparing it to the ground truth. Also, the system obtained a performance for glaucoma diagnosis that was better than 5 of 6 ophthalmologists, and this difference was statistically significant ($P < 0.05$). Furthermore, as demonstrated by Figures 3 and 4, there was no apparent biases within the appearance of photographs in the false-positive and false-negative groups for the deep learning system that would imply that deep learning system was systematically incapable of detecting certain features of glaucomatous optic neuropathy. In 4 of the Pegasus misclassifications (Figs. 1A, B, E, and F), the photographs were

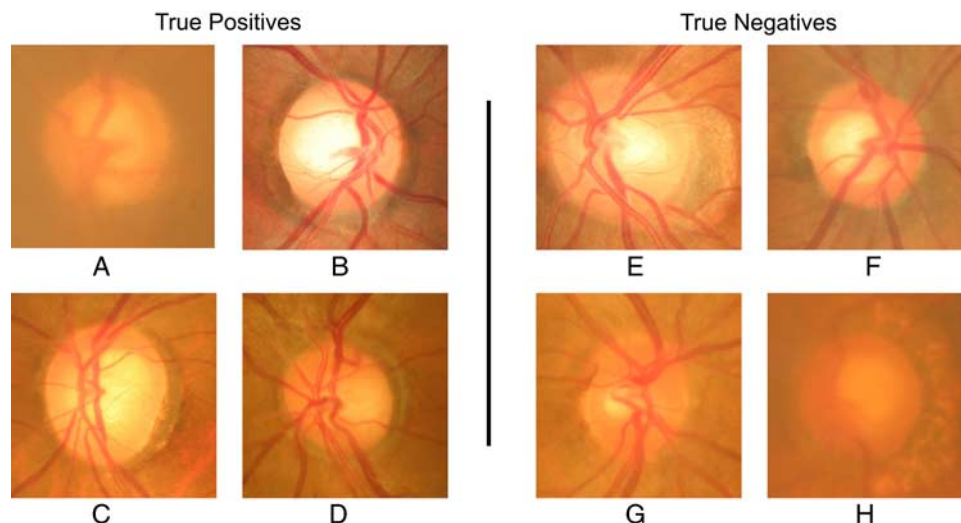


FIGURE 4. Randomly selected photographs that were classified correctly by the deep learning system. The true positives (A–D) were correctly identified as having glaucoma by the deep learning system, whereas the true negatives (E–H) were correctly identified as being nonglaucoma. The photographs have been manually cropped to just the optic disc region.

close to glaucoma/nonglaucoma decision boundary, therefore implying that the ophthalmologists were unclear on the correct diagnosis.

In Table S2 (Supplemental Digital Content 1, <http://links.lww.com/IJG/A292>), the decreasing AUROC with the level of training might be due to lack of knowledge bias by the ophthalmologist and fellows in training or systemic bias by the trained ophthalmologist.^{11,12} However, there was no statistically significant difference between the AUROC performance of the deep learning system and the “best case” consensus between ophthalmologists. Therefore, a larger sample of images is required for testing whether this difference is statistically significant.

Finally, the deep learning system showed perfect repeatability and an analysis time that was 10% of the reading time of ophthalmologists. These results suggest that AI systems based on deep learning, such as Pegasus, can play an important role in glaucoma screening.

This study demonstrated a high degree of intraobserver agreement that ranged from 62% to 97%. Moreover, the graders were more consistent with measuring VCDR than diagnosing glaucoma. This is similar to a study by Reus et al,¹² which demonstrated that the diagnostic accuracy of detecting glaucoma by ophthalmologists is 80.5%. Similarly, the intraphysician agreement between ophthalmologic examination in the clinic and through telemedicine was also 86%. Campbell et al¹³ demonstrated that there is poor agreement on the classification of retinopathy of prematurity even among expert graders. As suggested by Campbell et al,¹³ the discrepancy might be based on systemic biases in which some physicians tend to “over call” disease and others tend to “under call” a disease. However, these discrepancies might lead to differences in treatment of the patient based on the expert’s systemic bias.¹¹ Similarly, the agreement varied between the 6 ophthalmologists and between them and the gold standard based on disc photography. In practice, the diagnosis of glaucoma is multifactorial and includes intraocular pressure, corneal thickness, visual field, disc appearance, and optical coherence tomography in addition to longitudinal data and other

patient information. It might be that AI can outperform physicians when there is an established protocol as it follows a binary decision algorithm and often unable to incorporate ambiguity. Therefore, in the future, systems like these might play an important role in assisting clinicians with establishing normalcy in addition to detecting disease based on a large dataset to classify patient examinations with a high degree of accuracy and consistency.

Recent studies on the development of machine learning models for the detection of glaucoma have shown that deep learning is capable of obtaining high diagnostic performance, frequently obtaining above 90% in terms of AUROC performance.^{14–16} However, these studies have not previously compared the performance of a deployed deep learning system to the performance of ophthalmologists for the same patients, nor compared the speed at which a diagnosis outcome could be determined from the photograph. In addition, only 1 recent paper by Liu et al¹⁴ has evaluated performance on a dataset (albeit of 30 images) that was completely disjoint from the datasets used in algorithm development. A strength of this study was that the dataset was completely disjoint from the algorithm development process, and it was possible to compare the diagnostic performance of ophthalmologists, and their optimal consensus, to the performance of the deep learning system.

Other AI models for glaucoma have been developed to detect abnormalities using visual fields, fundus photography, and optical coherence tomography.^{17–19} In addition to disease detection, AI systems can be used to establish normal examinations in ophthalmology. Given the plethora of training datasets, deep learning algorithms can be trained to potentially learn “normal” ophthalmologic examinations. In addition, such AI systems can be used in the developing field of teleophthalmology.²⁰ Remote screening allows increased access for underserved populations and earlier detection of ophthalmic disease.²⁰

The strengths of this study include the graders with varying degrees of experience and who were masked to the analysis results of the deep learning system, the distribution of normal controls and glaucoma patients, and to the repeated images used for assessing intraobserver agreement

assessment. The limitations of this study include the standardization of the dataset, with the acquisition from the same fundus camera and field of view, as well as the relatively small sample size. The small sample size made it difficult to perform informative studies on the performance of the deep learning system on different subsets of the dataset, for example: is the performance of the deep learning system different when only examples of peripapillary atrophy are used? Future work will explore the performance of the deep learning system on images with different characteristics such as peripapillary atrophy.

The speed of interpretation and the lack of variability in deep learning systems such as Pegasus with its high sensitivity offer exciting possibilities in the development of scalable eye care programs to manage the exponentially rising need for eye care in the future.¹ Further investigations, in different settings, different datasets, and larger sample size, are required to further evaluate both the clinical and economic effects of such deep learning systems in the context of eye care.

REFERENCES

1. Tham Y-C, Li X, Wong TY, et al. Global prevalence of glaucoma and projections of glaucoma burden through 2040: a systematic review and meta-analysis. *Ophthalmology*. 2014;121:2081–2090.
2. Kapoor R, Walters S, Al-Aswad L. The current state of artificial intelligence in ophthalmology. *Surv Ophthalmol*. 2018;64:233–240.
3. Patel VL, Shortliffe EH, Stefanelli M, et al. The coming of age of artificial intelligence in medicine. *Artif Intell Med*. 2009;46:5–17.
4. DeepLearning4j. What's the difference between artificial intelligence (AI), machine learning and deep learning? open-source, distributed deep learning for the JVM. Available at: <https://deeplearning4j.org/ai-machinelearning-deeplearning>. Accessed June 6, 2018.
5. Wong TY. Prediction of diseases via ocular imaging: The Singapore Retinal Archival and Analysis Imaging Network, Inaugural Ocular Imaging Symposium; June 2008.
6. Zhang Z, Yin F, Liu J, et al. Origa-light: an online retinal fundus image database for glaucoma analysis and research. Engineering in Medicine and Biology Society (EMBC), 2010 Annual International Conference of the IEEE. August 31, 2010–September 4, 2010. Buenos Aires, Argentina: IEEE; 2010.
7. Rosman M, Zheng Y, Lamoureux E, et al. Review of key findings from the Singapore Malay Eye Study (SiMES-1). *Singapore Med J*. 2012;53:82–87.
8. He K, Zhang X, Ren S, et al. Deep residual learning for image recognition. *arXiv preprint*. 2015;arXiv:1512.03385.
9. Foong AWP, Saw S-M, Loo J-L, et al. Rationale and methodology for a population-based study of eye diseases in Malay people: The Singapore Malay Eye Study (SiMES). *Ophthalmic Epidemiol*. 2007;14:25–35.
10. DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44:837–845.
11. Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med*. 2016;15:155–163.
12. Reus NJ, Lemij HG, Garway-Heath DF, et al. Clinical assessment of stereoscopic optic disc photographs for glaucoma: The European Optic Disc Assessment Trial. *Ophthalmology*. 2010;117:717–723.
13. Campbell JP, Ryan MC, Lore E, et al. Diagnostic discrepancies in retinopathy of prematurity classification. *Ophthalmology*. 2016;123:1795–1801.
14. Liu S, Graham SL, Schulz A, et al. A deep learning-based algorithm identifies glaucomatous discs using monoscopic fundus photographs. *Ophthalmol Glaucoma*. 2018;1:15–22.
15. Christopher M, Belghith A, Bowd C, et al. Performance of deep learning architectures and transfer learning for detecting glaucomatous optic neuropathy in fundus photographs. *Sci Rep*. 2018;8:16685.
16. Shibata N, Tanito M, Mitsuhashi K, et al. Development of a deep residual learning algorithm to screen for glaucoma from fundus photography. *Sci Rep*. 2018;8:14665.
17. Muhammad H, Fuchs TJ, De Cuir N, et al. Hybrid deep learning on single wide-field optical coherence tomography scans accurately classifies glaucoma suspects. *J Glaucoma*. 2017;26:1086–1094.
18. Jae S, Cho KJ, Oh S. Development of machine learning models for diagnosis of glaucoma. *PLoS One*. 2017;2:e0177726.
19. Li Z, He Y, Keel S, et al. Efficacy of a deep learning system for detecting glaucomatous optic neuropathy based on color fundus photographs. *Ophthalmology*. 2018;125:1199–1206.
20. Rath S, Tsui E, Mehta N, et al. The current state of teleophthalmology in the United States. *Ophthalmology*. 2017;124:1729–1734.